

PROJECT DESIGN PHASE-II

TECHNOLOGY STACK (ARCHITECTURE & STACK)

Date	01 /10/2026
Team ID	SWTID-2026-2467
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	4 Marks

Technical Architecture

The architecture shows the application logic, cloud boundary, data storage, access controls, monitoring, and the ticket-classification workflow used by the proposed ServiceNow solution.

AUTO TICKET CLASSIFICATION

Technology Architecture & Stack | ServiceNow Flow Designer

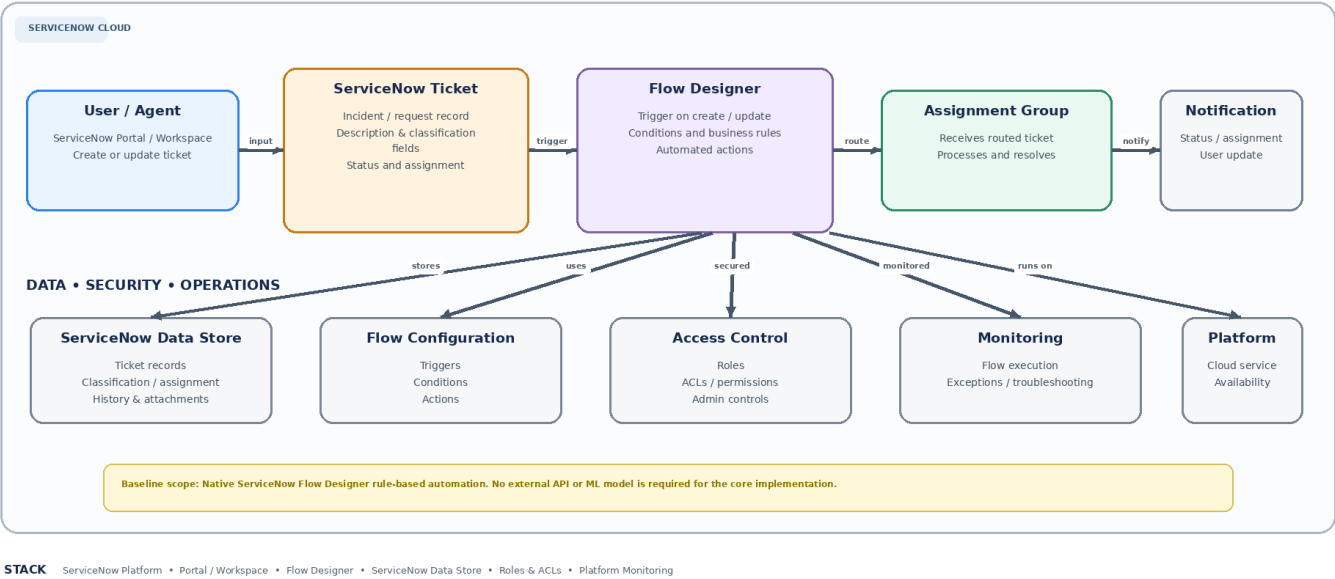


Figure 1. Technical architecture for Auto Ticket Classification using Flow Designer

Table 1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	How users interact with the application	ServiceNow Portal / ServiceNow Workspace
2	Application Logic-1	Ticket trigger, classification and routing workflow	ServiceNow Flow Designer

3	Application Logic-2	Business conditions used to determine classification	Flow Designer Conditions
4	Application Logic-3	Automated field updates and assignment routing	Flow Designer Actions
5	Database	Ticket records, fields, status and history	ServiceNow Data Store
6	Cloud Database	Cloud-hosted application data	ServiceNow-managed Cloud Platform
7	File Storage	Ticket attachments when required	ServiceNow Attachments / Platform Storage
8	External API-1	External third-party interface	Not required for baseline solution
9	External API-2	External third-party interface	Not required for baseline solution
10	Machine Learning Model	ML interface for classification	Not applicable — rule-based Flow Designer classification
11	Infrastructure (Server / Cloud)	Application execution and deployment	ServiceNow Cloud Platform

Table 2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	The baseline solution uses native ServiceNow capabilities; no separate open-source application framework is required.	ServiceNow Flow Designer
2	Security	Role-based access and ACLs restrict ticket and configuration access. Secure platform communication is used.	ServiceNow Roles & ACLs / HTTPS
3	Scalable Architecture	The solution can be extended with additional flows, conditions, categories and assignment groups as requirements grow.	ServiceNow Cloud / Flow Designer
4	Availability	The automation runs within the ServiceNow cloud platform; no separate application server is required for the baseline design.	ServiceNow Cloud Platform
5	Performance	Rule-based automation reduces manual processing. Conditions and actions are kept focused to support efficient execution.	Flow Designer

Architecture note: The current project scope is a rule-based ServiceNow automation. If ML-based classification is introduced in a later phase, a model/service interface can be added to the architecture without changing the overall ticket-routing concept.

References

<https://c4model.com/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>